

# Mean–Variance Portfolio Selection

FIN6131 Portfolio and Asset Management: Assignment 1

19 September 2026

## 1 Data, estimation, and portfolio theory

**Research question.** Does the in-sample benefit of allowing short sales survive a chronological holdout? Following the mean–variance framework of Markowitz [1], this study separates the attainable risk–return trade-off from the reliability of estimated optimal weights. The principal finding is that short selling improves the fitted frontier, but extreme leverage makes unrestricted tangency fragile out of sample.

**Sample and provenance.** Nine USD-denominated ETFs span US large-cap equities (SPY), Nasdaq-100 (QQQ), US small-cap equities (IWM), developed ex-US equities (EFA), emerging equities (EEM), long US Treasuries (TLT), investment-grade corporate bonds (LQD), gold (GLD), and US REITs (VNQ). The five-year return window is 19 September 2021 through 18 September 2026. Yahoo adjusted closes account for splits and distributions [3]; they serve as a historical total-return proxy. Direct `yfinance` access was blocked, so the same provider’s public chart responses were obtained through a browser on 19 September 2026. The executable notebook includes the documented `yfinance` download option [2] and a verified offline snapshot. Raw responses, exact URLs, and checksums are archived.

There are 1,256 aligned price observations per ETF, including the prior close of 17 September 2021, yielding **1,255 daily returns** from 20 September 2021 onward. There are no missing cells, duplicate dates, imputed prices, or dropped trading rows. For adjusted price  $P_{it}$ , estimation uses

$$r_{it} = \log(P_{it}/P_{i,t-1}), \quad \hat{\mu} = 252\bar{r}, \quad \hat{\Sigma} = \frac{252}{T-1} \sum_{t=1}^T (r_t - \bar{r})(r_t - \bar{r})^\top.$$

The sample covariance is positive definite (condition number 238.69). Full-sample annual log means range from  $-8.66\%$  for TLT to  $17.99\%$  for GLD. These are historical estimates, not forecasts. Asset statistics and the complete covariance matrix appear in the notebook.

**Optimization and the efficient set.** With  $\mathbf{1}^\top w = 1$ , expected return and variance are  $w^\top \hat{\mu}$  and  $w^\top \hat{\Sigma} w$ . For each target  $m$ , solve

$$\min_w \frac{1}{2} w^\top \hat{\Sigma} w \quad \text{subject to} \quad \mathbf{1}^\top w = 1, \quad w^\top \hat{\mu} = m, \quad \begin{cases} w \in \mathbb{R}^9 & (\text{U : shorts allowed}), \\ w_i \geq 0 & (\text{L : long-only}). \end{cases}$$

The global minimum variance portfolio (MVP) removes the target constraint; the efficient frontier is the upper branch starting at that regime’s MVP. The budget condition remains in the “unconstrained” case. With  $A = \mathbf{1}^\top \hat{\Sigma}^{-1} \mathbf{1}$ ,  $B = \mathbf{1}^\top \hat{\Sigma}^{-1} \hat{\mu}$ ,  $C = \hat{\mu}^\top \hat{\Sigma}^{-1} \hat{\mu}$ , and  $D = AC - B^2$ , unrestricted solutions satisfy

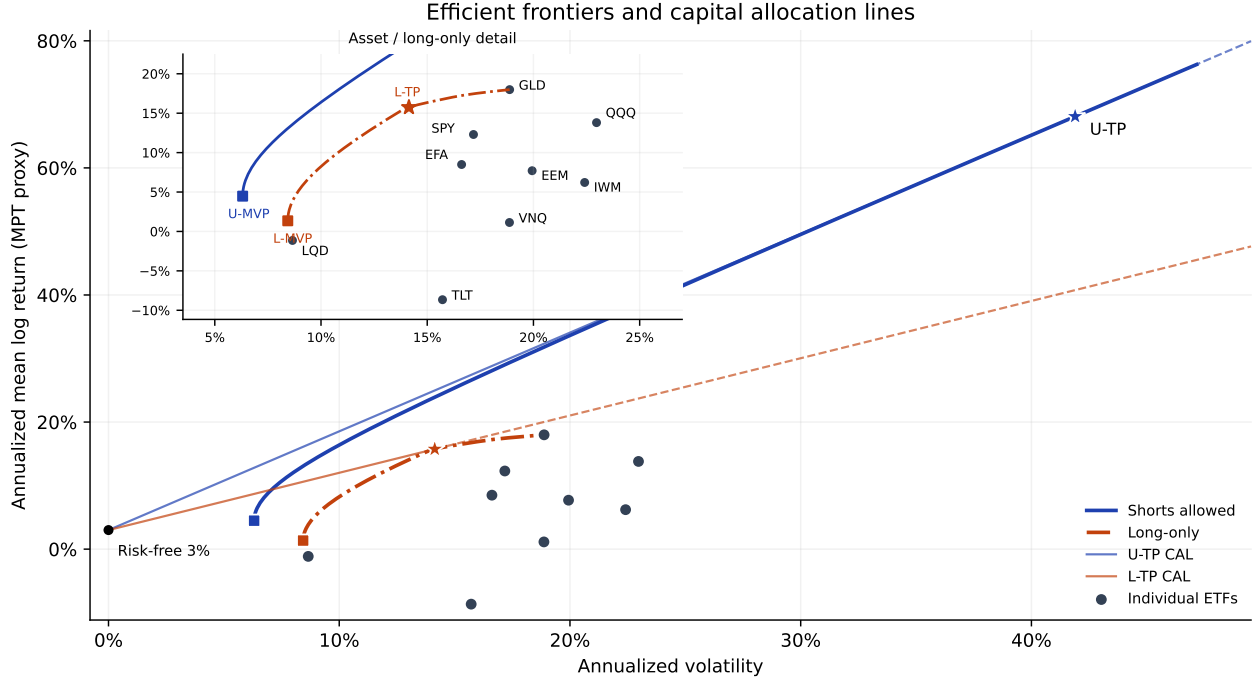
$$w_G = \frac{\hat{\Sigma}^{-1} \mathbf{1}}{A}, \quad w(m) = \hat{\Sigma}^{-1} \left( \frac{C - Bm}{D} \mathbf{1} + \frac{Am - B}{D} \hat{\mu} \right), \quad \sigma^2(m) = \frac{Am^2 - 2Bm + C}{D}.$$

Long-only portfolios are solved by checked SLSQP quadratic programs. Feasible-set inclusion implies that short-sale restrictions cannot lower minimum variance at a common target return. They also bound attainable means between the smallest and largest asset means; unrestricted target returns have no finite upper bound.

**Tangency and cash allocation.** At the stipulated annual  $r_f = 3\%$ , tangency maximizes  $S(w) = (w^\top \hat{\mu} - r_f)/\sqrt{w^\top \hat{\Sigma} w}$ . For  $e = \hat{\mu} - r_f \mathbf{1}$ ,  $w_T = \hat{\Sigma}^{-1} e / (\mathbf{1}^\top \hat{\Sigma}^{-1} e)$  in regime U; its normalization is positive in both estimation windows. For L, the convex program  $\min_{z \geq 0} z^\top \hat{\Sigma} z$  subject to  $e^\top z = 1$ , followed by  $w = z / (\mathbf{1}^\top z)$ , avoids a local ratio optimum. Allocation  $y$  to tangency and  $1 - y$  to cash gives the CAL  $\mu_C = r_f + S_T \sigma_C$  for  $y \geq 0$ . Lending corresponds to  $0 \leq y \leq 1$ ;  $y > 1$  assumes borrowing at the same rate. The risky-sleeve short-sale restriction does not itself forbid borrowing cash.

**Return-convention caveat.** As requested, these optimization formulas use annualized *asset log-return* moments and subtract 0.03 directly. Weighted asset log returns are not exact portfolio log returns: the frontier, Sharpe objective, and CAL are conventional MPT approximations, particularly under leverage. The holdout below instead compounds exact weighted simple returns. Converting 3% to  $\log(1.03)$  changes full-sample long-only tangency weights by at most 0.060 percentage points, but does not remove the broader approximation.

## 2 Full-sample frontier and economic interpretation



**Figure 1:** Both efficient frontiers, four optimized portfolios, individual ETFs, and both CALs. The inset labels assets and low-risk portfolios on the same axis definitions. Solid CAL segments involve lending; dashed extensions assume borrowing. The unrestricted frontier is shown over a finite range, not truncated economically.

**Table 1:** Full-sample weights (%).

| ETF | U-MVP  | L-MVP | U-TP    | L-TP  |
|-----|--------|-------|---------|-------|
| SPY | 54.21  | 6.47  | 519.43  | 39.42 |
| QQQ | -35.24 | 0.00  | -159.19 | 0.00  |
| IWM | -10.47 | 0.00  | -104.04 | 0.00  |
| EFA | -13.57 | 0.00  | -36.34  | 0.00  |
| EEM | 7.90   | 0.00  | -50.89  | 0.00  |
| TLT | -56.48 | 0.00  | -135.38 | 0.00  |
| LQD | 157.56 | 85.06 | -7.02   | 0.00  |
| GLD | 4.87   | 8.48  | 160.36  | 60.58 |
| VNQ | -8.78  | 0.00  | -86.92  | 0.00  |

**Table 2:** Annual fitted MPT statistics.

| Portfolio | $\mu$ (%) | $\sigma$ (%) | Sharpe | Gross ( <i>imes</i> ) |
|-----------|-----------|--------------|--------|-----------------------|
| U-MVP     | 4.48      | 6.31         | 0.234  | 3.491                 |
| L-MVP     | 1.35      | 8.44         | -0.196 | 1.000                 |
| U-TP      | 68.10     | 41.89        | 1.554  | 12.596                |
| L-TP      | 15.75     | 14.14        | 0.901  | 1.000                 |

U = shorts allowed; L = long-only; TP = tangency portfolio. Gross exposure is  $\sum_i |w_i|$  relative to net wealth. Displayed weights are rounded; unrounded weights sum to one. Zero weights are active nonnegativity constraints.

**Diversification versus leverage.** U-MVP achieves 6.31% volatility, compared with 8.44% for L-MVP and 8.66% for the least-volatile individual ETF, LQD. Covariance hedges therefore reduce risk below that of any single asset. The unrestricted solution uses a 157.56% LQD holding and a -56.48% TLT position, among other offsets; its 3.49-times gross exposure is not a low-leverage investment. The long-only MVP places 85.06% in LQD, sacrificing hedge flexibility. Its 1.35% fitted mean lies below cash's 3%: it minimizes *risky-asset variance*, not investor utility, and is dominated by cash once the risk-free asset is admitted.

**The price of the fitted Sharpe improvement.** U-TP raises fitted Sharpe from L-TP's 0.901 to 1.554, but requires 12.60-times gross exposure: 519.43% SPY and 160.36% GLD, financed by shorts in the other seven ETFs. Its 68.10% fitted mean accompanies 41.89% volatility, not an arbitrage. L-TP instead allocates 39.42% to SPY and 60.58% to GLD, with 15.75% fitted mean and 14.14% volatility. Holding many available ETFs is not necessarily optimal when their covariances overlap; nevertheless, a two-asset solution is exposed to concentration and sample-specific mean estimates. Restricting shorts lowers the fitted Sharpe by construction, while potentially reducing estimation-error amplification.

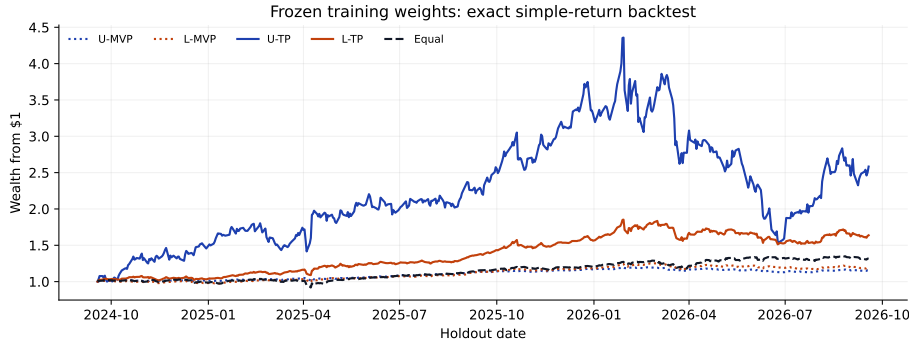
### 3 Chronological robustness and limitations

The first three calendar years contain **754 training returns** (through 18 September 2024); the last two contain **501 holdout returns** (19 September 2024–18 September 2026). All four optimizers are refitted using training data only. Those *target* weights remain frozen and are restored by daily rebalancing; equal weighting uses the identical convention with  $w_i = 1/9$ . This is not fixed-share buy-and-hold. Training L-TP holds 26.58% SPY and 73.42% GLD, whereas training U-TP has 14.68-times gross exposure. Full-sample weights never enter the holdout evaluation.

For each holdout date,  $R_{pt} = \sum_i w_i [\exp(r_{it}) - 1]$  and  $W_t = \prod_{s \leq t} (1 + R_{ps})$ . The first holdout return uses the final training close. Realized Sharpe is  $\sqrt{252}(\bar{R}_p - R_{f,d})/s(R_p)$ , with  $R_{f,d} = 1.03^{1/252} - 1$ ; CAGR is  $W_N^{252/N} - 1$ . Drawdown uses  $W_0 = 1$  in the running peak. Costs, taxes, margin calls, and stock-loan fees are omitted.

**Table 3:** Holdout results from frozen training weights; drawdowns are signed peak-to-trough losses.

| Portfolio | CAGR (%) | Vol. (%) | Sharpe | Max DD (%) | Total (%) |
|-----------|----------|----------|--------|------------|-----------|
| U-MVP     | 7.53     | 5.52     | 0.808  | -5.93      | 15.53     |
| L-MVP     | 9.19     | 8.27     | 0.747  | -6.72      | 19.10     |
| U-TP      | 61.18    | 56.83    | 1.075  | -64.99     | 158.32    |
| L-TP      | 28.15    | 18.98    | 1.247  | -18.38     | 63.75     |
| Equal     | 14.75    | 12.55    | 0.924  | -11.48     | 31.47     |



**Figure 2:** Exact compounded holdout wealth, with all strategies starting from one dollar.

**Interpretation.** L-TP delivers the highest holdout Sharpe, 1.247, versus 1.075 for U-TP and 0.924 for equal weighting. Its 28.15% CAGR exceeds the benchmark’s 14.75%, but volatility (18.98% versus 12.55%) and drawdown (18.38% versus 11.48% in magnitude) are also larger: it does not dominate equal weighting on every criterion. U-TP delivers the largest total gain, 158.32%, yet loses 64.99% from its running peak. Both MVPs reduce volatility and drawdown relative to equal weighting but have lower CAGR and Sharpe. Thus the optimizer’s objective matters: minimum variance is not maximum realized reward per unit risk.

**What the robustness check does and does not establish.** U-TP’s training Sharpe proxy of 1.817 falls to a realized holdout Sharpe of 1.075, alongside volatility rising from 43.89% to 56.83%; the comparison also changes from log-moment to simple-return measurement. Long-only tangency wins on holdout Sharpe in this sample, consistent with constraints limiting unstable offsetting bets, but this one split cannot establish causality or statistical superiority. Historical means are noisy; covariances vary across regimes; overlapping equity exposures amplify estimation sensitivity. Surviving ETF selection, revised adjusted-price histories, fat tails, and omitted shorting/turnover costs further limit implementability. Future work should test rolling holdouts, mean/covariance shrinkage, leverage caps, and realistic financing and rebalancing costs rather than tune on this holdout.

**Reproducibility.** The standalone notebook embeds the checked CSV and all numerical functions; seed 6131 and locked dependencies support reruns. All 98 tests pass, including isolated offline notebook execution. Independent SLSQP and analytic weights agree within  $3.23 \times 10^{-7}$ ; common-target frontier dominance and a holdout-perturbation no-lookahead check pass. Tables and plots are generated from the same executed results. These are historical research results, not investment recommendations.

## References

- [1] H. Markowitz (1952). Portfolio Selection. *The Journal of Finance*, 7(1), 77–91. doi:10.1111/j.1540-6261.1952.tb01525.x.
- [2] yfinance. *download API documentation*. <https://ranaroussi.github.io/yfinance/reference/api/yfinance.download.html> (accessed 19 September 2026).
- [3] Yahoo Finance. *What is the adjusted close?* <https://help.yahoo.com/kb/SLN28256.html> (accessed 19 September 2026). Raw data provenance: repository `assignment1/data/metadata.json`.